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Palaeomagnetism of the Canary Islands and Madeira

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Summary

Palaeomagnetic results are presented for 1144 specimens from a total of 359 igneous bodies from the Canary Islands and Madeira. Available stratigraphic control, together with 90 K:Ar age determinations enable the data to be reliably assigned to the Pleistocene (t = 0–1·6 My); Pliocene (t = 1·6–5·1 My); and Miocene (t = 5·2–25 My) periods. The mean virtual geomagnetic pole positions for each of these periods are indistinguishable and are significantly different from the present geographic pole in a sense consistent with Wilson's offset dipole hypothesis. If considered as part of a polar wander path for the Tertiary era, the results suggest that if any marked change of polar wander rate occurred during the Mid-Tertiary, a slowing (rather than an increase) in the rate is likely to have occurred.

Introduction

The path defined by successive positions of geographic pole with respect to a given stable region is called the polar wander path. Comparison of such polar wandering paths for separated continental blocks provided initial important confirmation of continental drift (Creer, Irving & Runcorn 1957). Polar wander paths may also be significant from a different viewpoint: the possibility has been proposed by Irving & Robertson (1969) that periods of rapid movement of the pole may be related to periods of increased tectonic activity, or diastrophic intervals. At the same time, these authors believe that the polarity reversal frequency may also be related to tectonic maxima. Hide (1967) suggests that the mechanism of such a relationship could be due to irregularities in the core–mantle boundary simultaneously effecting both geomagnetic activity and upper mantle motions. Irving & Robertson (1969) suggest that according to available palaeomagnetic pole positions, the most recent period of rapid polar wandering was at about t = 25 My. They note that Vine (1966) suspects the existence of a change in the reversal frequency at the same time. In contrast, McElhinny & Wellman (1969) believe that comparison of apparent polar wander curves for the Cenozoic of Africa, Europe, and Australia show (after correction for continental drift) a uniform rate of polar wandering of 0·45° per My (or 5 cm per year), without any such period of rapid movement at t = 25 My as preferred by Irving & Robertson (1969).

It is the purpose of this communication to present a large volume of palaeomagnetic data from the Canary Islands and Madeira. A high degree of age control, resulting from potassium–argon dates on 90 different igneous bodies, together with

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previously published stratigraphic maps, facilitates reliable assignment of all data to the Pleistocene, Pliocene, or Miocene periods. These palaeomagnetic data will thus provide a unique opportunity to monitor the polar wandering rate for the past 25 My, relative to a restricted part of the Earth's crust. At the same time, the results are relevent to those hypotheses involving long-term distortion of the geomagnetic field away from that of a centred axial dipole (Wilson 1970; Watkins 1972).

Geology of the Canary Islands and Madeira

The Canary Islands and Madeira (Fig. 1) are almost completely volcanic in outcrop. The regional genesis and geology of each island are fully reviewed in Watkins, Richardson & Mason (1966a, b); and Abdel-Monem, Watkins & Gast (1971, 1972). These latter two publications, together with the work of Watkins & Abdel-Monem (1971) provide fine details of stratigraphic horizons and sample sites for all the units which were subject to K:Ar analyses.

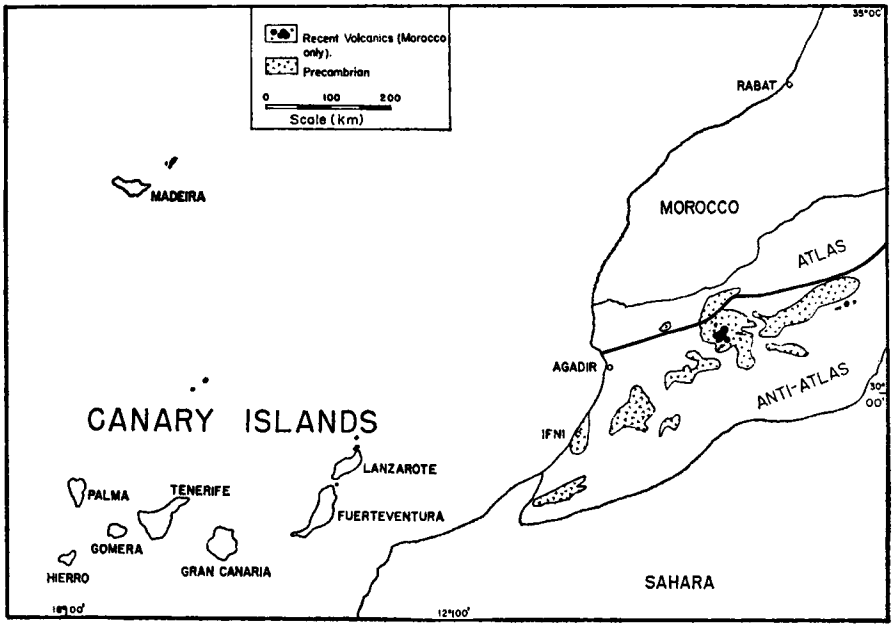


FIG. 1. Map showing the location of the seven major Canary Islands and Madeira.

Methods

Samples were collected during three field seasons (1965, 1966 and 1968) using gasoline-powered portable drills. Cores of 2·5-cm diameter were oriented in geographic co-ordinates while still attached to outcrop, to an accuracy of $\pm 3°$. The number of separate cores per body varied from 2 to 8, and averaged 4.

Palaeomagnetic properties of specimens (of 2·2-cm length) from all cores were determined using astatic and spinner magnetometers. All specimens were demagnetized in alternating magnetic fields of 100, 200, and 300 Oe, and mean directions of stable remanent magnetism were computed using minimum dispersion computation methods (see Watkins, Hajash & Abranson 1972 for discussion).

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Table 1

Palaeomagnetic data for each unit

(a) Pleistocene

Unit	N	D	I	J	$(-E)$	R	θ'	ϕ'	K:Ar(My)
(i) Lanzarote									
15	2	356·2	+51·9	0·180	3	1·966	85·9	294·6	—
16	2	000·4	+40·0	0·160	3	1·999	83·6	163·0	—
17	2	018·4	+46·1	0·313	3	1·945	73·7	78·0	—
18	2	348·1	+26·4	0·101	3	1·884	71·3	205·0	—
19	2	352·4	+34·9	0·230	3	1·999	77·9	203·3	—
21	2	009·3	+34·5	0·210	3	1·898	76·8	124·8	—
22	2	021·0	+38·0	0·100	3	1·960	69·5	94·0	—
23	2	22·7	+30·5	0·190	3	1·959	65·6	103·0	—
24	2	25·6	+32·4	0·581	4	1·978	63·9	97·1	—
25	2	27·8	+30·4	0·105	3	1·898	61·4	96·9	—
26	2	24·2	+36·3	0·204	3	1·840	66·3	93·3	—
27	2	28·6	+32·7	0·163	3	1·999	61·5	93·6	—
28	2	26·2	+31·5	0·205	3	1·959	63·1	97·5	—
29	2	28·6	+34·5	0·210	3	1·980	62·1	91·4	—
30	2	3·4	+32·0	8·000	4	1·990	79·0	149·9	—
31	2	26·7	+34·6	0·907	4	1·999	63·7	93·1	—
32	2	31·7	+35·7	0·800	4	1·965	59·8	87·3	—
33	2	24·6	+30·5	0·703	4	1·876	64·1	100·5	—
34	2	175·0	−16·5	1·000	3	1·980	−68·8	0·3	—
35	2	156·6	−20·2	1·100	3	1·950	−61·2	40·8	—
36	2	168·9	−22·0	2·600	3	1·888	−69·5	19·5	—
37	2	171·9	−13·8	1·200	3	1·965	−66·6	7·1	—
38	2	176·3	−21·9	9·800	3	1·868	−71·9	358·1	—
39	2	182·7	−24·9	1·430	2	1·994	−73·7	337·2	—
40	2	180·8	−27·7	6·000	3	1·884	−75·5	343·6	—
41	7	8·2	+45·9	6·760	3	6·988	82·6	91·1	0·03±0·03
42	6	32·0	+39·0	2·468	3	5·980	60·4	86·2	0·96±0·10
43	2	14·8	+61·2	4·129	3	2·000	72·1	28·1	—
46	5	22·1	+33·4	7·328	3	4·961	67·1	103·1	—
(ii) Gran Canaria									
28	2	3·4	+32·0	0·207	4	1·965	79·0	146·9	—
29	2	12·9	−06·9	0·300	4	1·990	56·3	140·7	—
30	2	44·9	+16·1	0·300	4	1·914	43·3	90·4	—
31	2	59·6	+05·2	0·101	4	1·850	27·9	87·1	—
32	2	354·9	+40·3	0·300	4	1·999	83·2	208·4	—
(iii) Tenerife									
7	4	348·6	+38·6	8·612	4	3·993	78·0	228·7	—
8	4	352·6	+26·1	6·048	4	3·987	74·2	194·5	—
9	5	343·1	+37·3	2·972	4	4·991	73·0	236·0	0·67±0·01
10	6	175·3	−23·4	1·747	4	5·485	−73·6	3·7	—
11	6	21·5	−45·2	2·728	4	5·372	31·5	144·6	0·84±0·28
12	8	165·3	−37·6	1·675	5	7·530	−74·9	52·9	1·35±0·02
12A	5	335·7	+38·9	7·977	4	4·977	67·2	247·3	1·03±0·05
12B	4	344·7	+32·5	1·019	3	3·995	72·5	224·2	—
13	4	6·0	+37·6	3·377	4	3·968	81·2	128·0	—
14	4	0·3	+40·6	1·803	3	3·990	85·2	163·8	—
15	4	240·2	−21·6	4·432	4	3·951	−31·5	261·1	—
16	6	10·8	+59·0	7·181	4	5·950	75·2	21·8	0·53±0·03
17	3	345·8	+44·7	2·507	3	2·987	77·3	252·9	—
18	6	354·4	+35·5	1·549	3	5·964	80·2	199·9	—
19	6	192·2	−38·8	4·212	4	5·901	−77·4	283·3	—
45	6	29·2	+28·8	1·328	3	5·908	60·1	96·1	—
46	3	46·5	+23·3	2·931	4	2·961	43·8	87·6	—
47	5	4·0	+51·9	5·482	4	4·959	84·3	23·4	—
48	6	4·6	+39·9	6·483	3	5·974	83·3	128·1	—
49	7	137·0	−41·4	1·015	3	6·617	−51·3	81·0	1·63±0·11
50	2	187·8	−31·1	4·500	2	1·989	−76·4	309·7	0·81±0·09
51	2	183·5	−47·5	4·600	3	1·999	−86·9	246·9	—
52	2	174·7	−37·1	5·700	3	1·998	−81·1	17·1	—
53	2	181·6	−41·9	7·900	3	1·956	−85·7	324·4	—
54	2	185·3	−42·6	9·200	3	1·998	−84·0	289·5	—
55	2	190·3	−44·1	6·900	3	1·965	−80·5	265·9	—
56	2	191·2	−48·1	7·500	3	1·988	−80·1	246·0	—
57	2	188·7	−48·7	1·390	2	1·902	−82·3	241·3	—
58	2	186·3	−45·0	1·040	2	1·865	−84·2	269·2	—
59	2	179·5	−53·8	4·800	3	1·986	−84·0	160·0	—
60	2	182·8	−48·0	1·050	2	1·910	−87·4	236·1	—
61	4	191·6	−41·9	1·541	3	3·977	−78·9	275·0	1·52±0·15
63	2	185·0	−40·4	2·500	3	1·985	−83·1	302·0	—
64	2	169·0	−34·9	3·100	3	1·869	−76·5	34·0	—
65	2	182·5	−12·7	3·300	3	1·945	−68·0	337·0	—
66	2	180·5	−17·8	7·200	3	1·995	−70·9	342·1	—
67	2	187·4	−15·1	2·200	3	1·869	−68·3	343·3	—
68	2	194·2	−32·8	9·000	3	1·928	−73·3	289·3	0·84±0·05
69A	2	207·0	−63·4	1·760	3	1·999	−62·7	211·7	0·86±0·22
70	4	238·9	−10·2	1·817	2	3·853	−29·7	267·9	—
71	6	353·4	43·8	2·949	4	5·963	83·6	236·5	1·54±0·17
71A	3	173·4	−26·6	6·645	4	2·991	−74·8	12·2	0·88±0·05
71B	4	166·6	−26·6	9·210	4	3·992	−71·3	31·6	—
71C*	4	223·2	−52·3	5·267	4	3·991	−52·7	238·6	—
72	4	339·7	+50·8	5·046	4	3·993	72·1	273·3	—
73	6	6·9	+47·7	2·400	4	5·969	83·8	67·8	—
74	3	16·3	+33·1	2·417	4	2·999	72·0	107·3	0·95±0·06

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Unit	N	D	I	J	$(-E)$	R	θ'	ϕ'	K:Ar(My)
75	2	282·7	+12·5	7·000	4	1·868	14·2	253·1	—
76	4	356·6	−14·3	2·781	3	3·876	54·6	173·0	—
77	2	12·3	+30·0	0·640	3	1·895	73·4	117·6	—
78	2	4·0	+23·6	0·878	3	1·990	73·7	149·6	—
79	3	6·1	+28·6	6·321	4	2·985	76·0	142·0	0·53±0·03
80	2	351·7	+36·5	0·320	3	1·999	63·9	182·7	—
81	2	344·2	+06·1	0·105	3	1·986	60·7	197·4	—
82	2	11·9	+30·2	0·400	3	1·995	73·8	118·4	—
83	3	184·7	−59·2	3·688	4	2·999	−77·4	183·9	1·77±0·04
83A	2	181·4	−40·8	2·600	3	1·985	−85·0	328·0	—
84	2	72·1	−82·1	2·000	4	1·694	−22·5	147·6	1·22±0·04
85	2	9·0	+23·1	1·358	3	1·995	71·9	137·5	0·20±0·08
86	5	339·0	+31·4	1·460	4	4·861	67·7	232·0	0·67±0·01
87*	4	183·7	−37·8	4·536	4	3·996	−82·4	320·1	—
88	4	19·5	+21·0	1·186	5	3·773	65·0	116·6	<0·20
(iv) Hierro									
1	2	353·9	+67·2	0·120	4	1·999	67·4	331·8	—
2	2	356·2	+10·1	8·601	4	1·986	67·1	171·8	0·19±0·03
3	2	48·8	+57·0	4·603	4	1·984	48·2	45·4	—
4	2	12·6	+52·6	0·190	3	1·950	77·8	42·0	0·70±0·07
5	2	20·4	+57·1	1·148	3	1·998	70·2	36·4	—
6	2	353·8	+64·9	5·162	4	2·000	70·2	329·3	—
7	2	349·5	+32·5	0·777	4	1·999	76·0	207·7	—
8	2	353·0	+50·1	3·444	4	1·998	83·1	281·0	—
14A	6	22·1	+6·2	1·076	3	5·951	57·6	117·4	—
16	2	16·1	+52·1	6·134	4	1·978	75·2	48·4	—
17	2	35·9	+16·5	4·201	4	2·000	51·0	94·6	—
19	2	31·5	+13·8	2·054	4	1·996	53·7	100·8	—
19	2	20·3	+18·9	3·783	4	1·999	63·7	111·2	0·19±0·01
20	2	335·9	+67·4	3·366	4	1·997	60·9	309·5	—
20A	4	26·7	+54·0	5·181	4	3·964	66·3	48·5	—
20B	6	8·7	+51·4	6·584	4	5·940	81·3	39·8	—
20C	4	10·2	+37·0	1·249	3	3·982	78·3	106·8	—
20D	4	9·1	+46·7	2·282	3	3·999	82·0	67·8	—
20E	3	7·7	+38·4	8·246	4	2·949	80·7	111·0	—
21	4	355·8	+46·9	1·551	3	3·975	86·3	259·3	—
21A	8	326·1	+53·1	1·121	3	7·843	60·4	272·1	—
22	2	340·8	+44·8	0·204	3	1·989	72·9	252·0	—
23	2	0·6	+43·4	0·103	3	1·995	87·4	148·9	0·74±0·04
24	2	334·3	+49·1	0·504	3	1·905	67·4	264·0	—
25	2	346·2	+39·4	0·104	3	1·898	76·4	231·5	—
27	2	18·9	+41·0	9·392	4	2·000	72·4	81·6	—
28	4	346·5	+51·9	6·660	4	3·899	77·4	277·5	—
29	2	38·0	+44·1	8·906	4	1·844	56·1	66·1	—
29A	6	1·1	+53·3	1·139	3	5·990	83·7	350·1	—
30	2	10·3	+49·0	1·956	3	1·992	80·7	55·5	—
31	2	10·0	+41·6	9·283	4	1·998	80·2	92·5	—
32	2	21·7	+57·3	9·616	4	1·999	69·2	37·0	—
33	2	17·4	+63·9	1·806	3	1·999	67·5	15·0	—
33A	4	7·0	+56·7	1·024	3	3·992	78·8	11·8	—
34	2	41·5	+57·7	6·113	4	1·911	54·0	44·0	—
(v) La Palma									
1	4	165·6	−10·4	1·063	3	3·352	−62·9	18·7	1·57±0·09
2	7	163·2	−37·9	1·808	3	6·878	−73·1	53·7	1·02±0·03
3	4	158·4	−30·6	1·966	3	3·720	−66·7	48·9	—
4	5	21·5	+47·1	4·031	3	4·983	71·1	71·9	—
5	6	332·7	+55·3	2·175	4	3·974	66·0	280·1	—
6	4	359·1	+42·8	1·702	3	3·999	86·0	177·7	—
7	3	340·9	+52·3	3·913	3	2·878	73·1	275·0	—
8	5	40·1	+47·5	1·056	3	4·987	55·0	66·0	—
9*	6	0·5	+29·9	1·192	3	5·902	77·3	163·4	0·60±0·30
10	7	0·8	+38·4	3·347	3	6·982	82·9	159·5	—
11	4	2·8	+39·8	3·561	3	3·988	83·4	143·0	—
12	6	342·9	+45·6	9·528	4	5·924	74·8	253·5	—
13	6	344·8	+33·2	2·372	3	5·983	72·5	222·0	—
14	5	352·1	+10·1	5·484	4	4·348	65·2	185·0	—
15	4	12·7	+23·5	2·358	3	3·989	69·8	127·5	—
16	6	15·1	+38·8	6·219	4	5·766	74·8	99·1	—
17	4	166·3	−46·4	2·479	3	3·997	−77·9	74·3	0·81±0·09
18	6	3·1	+50·4	1·259	3	5·823	86·4	31·9	0·76±0·09
19	5	214·7	+1·0	5·310	4	4·813	−45·8	291·0	—
20	4	7·4	+48·4	1·811	3	3·981	83·5	68·2	—
21	6	0·5	+32·0	2·957	3	5·981	78·6	163·3	—
(vi) Madeira									
22	2	1·5	+45·8	2·600	4	1·997	84·3	149·1	—
23	2	351·2	+49·3	1·808	3	2·000	82·1	236·5	1·15±0·06
24	6	8·4	+55·1	5·314	4	5·920	82·5	47·6	0·89±0·05
25	2	159·5	−38·7	6·220	4	1·982	−68·9	47·3	1·03±0·06
26	8	185·6	−50·4	6·957	4	7·883	−85·0	269·8	1·03±0·06
27	8	170·4	−55·9	7·015	4	7·556	−81·2	101·1	1·05±0·06
29	4	249·0	−56·4	1·331	3	3·883	−32·3	230·1	0·74±0·04
30	6	7·0	+41·5	2·873	3	5·972	79·2	126·4	1·64±0·03
				(b) Pliocene					
i) Fuerteventura									
13	6	148·4	−54·0	2·392	3	5·963	−62·5	100·4	4·25±0·44
17*	5	239·4	−68·1	1·185	5	4·296	−40·6	215·0	—
18	6	188·9	−31·8	2·931	3	5·977	−76·1	311·4	1·83±0·24

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Unit	N	D	I	J	$(-E)$	R	θ'	ϕ'	K:Ar(My)
(ii) Gran Canaria									
25	3	19·2	+45·2	2·767	3	2·923	72·9	77·9	—
25A	3	29·0	+14·1	7·858	5	2·311	55·6	109·8	3·75±0·12
25B	3	359·2	+29·2	1·015	3	2·998	77·6	171·5	—
29	2	34·3	+26·5	4·800	3	1·995	55·2	90·9	—
30	2	6·0	+29·8	2·600	3	1·986	76·8	138·2	—
31	2	9·0	+30·8	5·700	3	1·996	75·9	126·5	—
32	2	8·0	+24·8	9·200	3	1·908	73·2	136·2	—
34	7	185·5	−12·9	7·357	5	5·624	−67·9	333·4	2·80±0·08
35	7	195·8	−30·7	1·142	3	6·953	−71·4	292·9	2·20±0·11
36	3	199·9	−13·8	1·221	3	2·288	−61·8	302·5	—
37*	7	172·4	−57·7	7·337	4	6·913	−77·9	138·6	1·98±0·10
38	3	184·2	−48·5	9·712	3	2·980	−86·1	235·6	—
39	3	191·2	−29·5	3·106	3	2·926	−74·0	305·4	2·05±0·32
40	2	354·9	+40·3	9·000	4	1·998	83·2	208·4	—
40A	4	338·5	+40·5	1·947	4	3·287	70·1	248·9	—
41	3	172·7	−4·5	4·104	4	2·922	−63·3	4·5	3·50±0·09
59	2	175·7	−32·9	2·200	3	1·990	−79·1	6·8	—
(iii) Tenerife									
20*	6	219·4	−58·7	9·727	6	5·070	−55·6	227·5	—
26	2	192·4	−43·8	4·100	3	1·965	−78·6	265·5	—
27	3	196·3	−26·4	1·248	4	2·778	−69·3	296·6	4·65±0·70
28	2	186·2	−37·8	2·300	3	1·925	−80·8	304·8	—
29	2	185·5	−29·7	1·800	3	1·985	−76·4	320·6	—
30	2	178·0	−37·5	1·700	3	1·999	−82·3	357·6	—
31	2	172·8	−25·6	3·000	3	1·965	−73·5	9·3	—
32	5	176·5	−23·5	1·583	3	4·994	−73·9	359·5	—
33*	6	170·5	−33·3	1·303	3	5·981	−76·9	30·7	—
34	4	181·9	−26·5	7·327	4	3·981	−75·9	339·6	—
35	5	170·6	−34·0	2·970	3	4·746	−77·3	31·9	—
37*	3	175·1	−41·5	6·674	5	2·949	−84·0	34·8	—
38	2	173·0	−52·1	2·000	3	1·905	−82·7	110·4	—
39	2	13·6	+52·6	3·000	3	1·999	77·4	48·4	—
40	2	22·9	+55·5	1·800	3	1·915	69·3	46·5	—
41	2	10·1	+48·8	1·000	3	1·925	81·1	63·1	—
42	2	205·9	−32·1	3·000	3	1·948	−63·8	273·2	—
43	2	215·9	−15·3	9·000	4	1·864	−50·4	278·2	—
44	2	185·0	−38·1	1·100	3	1·985	−81·6	309·8	—
62	2	166·5	−51·4	1·833	3	2·000	−77·6	99·4	1·85±0·37
(iv) La Gomera									
4	6	15·0	+34·2	2·043	3	5·993	73·4	104·0	4·69±0·12
5	5	174·2	−34·4	2·500	3	4·969	−79·3	13·5	—
6*	4	216·2	+63·0	6·263	3	3·922	−10·2	317·3	—
7*	4	335·0	+24·0	8·965	3	3·682	61·9	223·9	—
(v) Hierro									
9	2	10·2	+36·1	6·254	4	1·955	77·9	109·0	—
10	2	19·8	+64·9	1·007	3	1·999	65·3	15·6	—
11	3	22·2	+47·3	1·865	2	2·994	70·4	64·4	—
12	2	351·0	+55·6	2·272	3	1·982	78·7	302·2	—
13	2	16·4	+27·8	6·949	4	1·997	70·0	108·9	—
14	2	354·9	+41·7	1·493	3	1·999	84·1	214·0	—
15	2	5·1	+37·7	4·010	4	1·998	81·9	125·5	—
15A	8	11·2	+43·8	1·355	3	7·915	79·8	81·1	3·05±3·00
(vi) Madeira									
1	2	183·2	−56·2	1·207	3	1·999	−85·2	194·6	—
2	2	193·8	−47·8	1·202	3	1·993	−77·5	267·5	—
3	2	166·9	−52·6	5·076	4	1·998	−79·0	79·1	—
4	2	185·8	−42·5	1·835	3	1·993	−80·4	309·5	—
5	2	180·8	−48·1	5·668	4	1·993	−86·3	332·2	—
6	2	174·4	−41·7	3·752	4	1·994	−80·0	13·7	—
7	2	190·2	−53·7	1·520	4	1·999	−81·3	239·9	—
8	2	190·0	−50·3	4·321	3	2·000	−81·4	261·4	—
9	2	221·6	−54·3	6·113	4	1·997	−55·6	237·8	—
10	2	165·3	−27·4	4·978	4	2·000	−67·4	22·9	—
11	2	189·4	−39·2	3·037	4	2·000	−76·6	302·5	—
12	2	194·2	−41·6	1·927	4	1·999	−74·8	284·5	—
13	2	182·2	−50·9	1·229	3	1·997	−87·8	281·7	—
14	2	183·1	−39·6	2·013	3	2·000	−79·4	327·1	—
15	2	170·3	−40·2	2·312	3	1·998	−77·0	26·5	—
16	2	182·9	−29·5	5·895	4	1·990	−72·9	333·5	—
17	2	183·3	−56·5	5·571	4	1·996	−84·8	193·4	3·05±0·15
18	2	156·0	−47·7	3·069	4	1·981	−69·1	68·8	—
19	2	169·3	−40·3	4·484	4	1·998	−76·5	29·8	—
20	2	181·4	−44·3	1·530	3	1·999	−83·2	332·1	—
21	2	357·1	51·6	1·856	3	1·999	87·5	243·3	—
28	8	184·3	−39·3	9·936	4	7·991	−78·9	321·8	1·76±0·07
(c) Miocene									
(i) Lanazrote									
1	2	182·7	−29·6	6·000	3	1·999	−76·3	335·5	—
2	2	209·6	−33·5	6·200	3	1·909	−60·9	272·0	—
3	5	205·6	−27·7	4·242	3	4·987	−62·4	285·7	5·82±0·36
4	6	352·5	+23·8	7·926	4	5·981	72·1	194·3	9·25±0·57
5	4	28·7	+51·5	1·747	3	3·929	65·2	65·4	6·00±0·15
6	2	14·3	+54·0	1·170	2	1·900	76·8	49·3	—
7	2	29·8	+42·8	1·900	3	1·985	63·2	78·8	—
8	2	22·8	+66·8	9·500	3	1·990	63·4	20·8	—
9	2	25·7	+38·5	5·600	3	1·980	65·7	88·5	—
10	2	16·0	+37·1	4·700	3	1·950	73·2	103·2	—
11	2	23·1	+34·9	1·200	3	1·985	66·8	96·9	—
12	2	25·8	+38·5	3·000	4	1·905	65·6	88·5	5·30±0·19

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Unit	N	D	I	J	(−E)	R	θ′	φ′	K:Ar(My)
14	6	54·4	+9·2	1·837	3	5·721	33·2	94·6	7·00±0·34
44	8	274·6	+79·0	5·835	4	7·325	28·5	325·8	7·68±0·53
45	8	10·9	+46·3	8·274	4	7·683	80·4	85·5	19·00±0·68
47	8	207·8	−13·8	3·127	3	7·962	−55·8	294·6	—
48	4	190·1	−44·9	3·018	3	3·984	−80·7	273·5	6·37±0·13
(ii) Fuerteventura									
1	3	0·8	+27·5	2·879	4	2·765	76·2	166·5	14·30±0·52
2	5	96·8	+66·5	1·010	4	4·848	16·9	32·5	15·88±1·68
3	5	7·3	+39·9	4·262	4	4·843	81·3	119·2	—
4	4	18·1	+47·8	2·696	4	3·946	74·1	73·6	16·55±0·61
5	4	171·9	−26·7	1·657	3	3·996	−73·8	19·2	—
6	4	97·7	−18·5	4·118	4	3·730	−11·2	84·4	—
7	5	115·0	−22·5	6·654	4	4·384	−27·4	78·4	—
8*	3	226·6	+12·6	1·478	4	2·991	−33·2	289·9	—
12*	4	142·6	−3·4	1·605	5	3·710	−45·5	49·5	19·60±0·42
15*	3	301·3	−46·2	2·385	5	2·883	10·7	40·0	—
16	6	197·4	−44·7	5·219	4	5·769	−74·4	263·2	—
19	3	27·5	+24·6	8·523	4	2·943	60·1	105·2	—
20	6	184·1	−65·1	7·120	4	5·896	−71·0	178·2	20·60±0·94
21	2	279·7	+27·6	7·450	4	—	15·3	268·2	—
22	4	189·4	−47·8	8·339	4	3·909	−81·8	254·1	—
23	4	190·0	−33·4	8·097	4	3·959	−76·3	305·4	11·80±0·33
1	2	315·8	+53·7	8·997	5	1·945	52·0	278·8	15·91±0·33
2	2	329·1	±62·0	8·522	4	1·995	60·9	297·8	—
3	2	6·8	+51·3	2·229	3	2·000	82·9	42·6	13·10±0·50
4	3	355·2	+49·5	7·187	3	2·924	85·2	288·1	15·0±0·38
5	2	351·3	+66·4	1·100	3	1·993	68·1	328·8	—
6	2	4·0	+66·3	1·800	3	1·898	69·1	351·6	—
7	2	351·4	+38·4	6·010	4	1·964	79·9	216·8	—
8	2	182·2	+75·9	1·181	3	1·994	01·4	343·2	—
9	2	358·3	+51·8	2·700	3	1·886	85·3	326·6	—
10	2	356·4	−78·4	2·659	3	1·995	−05·8	165·6	—
11	2	0·4	+48·6	1·114	3	1·986	88·4	357·8	—
12	2	357·9	+4·6	5·286	4	1·989	64·2	169·1	—
13	2	357·6	+40·8	4·511	4	1·898	84·8	189·8	—
14	2	348·2	+37·0	5·073	4	1·998	77·0	222·3	—
15	2	359·9	+38·4	5·106	4	1·980	83·6	165·0	—
16	2	345·2	+44·8	4·401	3	1·800	76·8	250·6	—
17	2	14·8	+47·9	9·342	4	1·999	77·0	66·5	—
18	2	21·0	−13·1	2·113	4	1·950	49·8	130·7	—
19	2	276·3	+5·4	1·064	3	1·698	6·9	253·6	—
(iii) Gran Canaria									
21	2	358·8	+46·5	1·504	3	1·990	89·0	243·2	—
22	4	20·4	+47·1	1·598	2	3·924	72·0	72·4	15·4±0·26
23	2	40·3	+39·4	1·277	3	1·985	53·3	73·9	—
24	3	348·7	+34·3	1·467	2	2·620	76·2	218·8	—
26	2	180·5	−29·6	2·422	3	1·998	−77·9	342·2	—
27	2	221·8	−33·6	1·025	3	1·895	−50·6	262·9	—
33	3	94·0	+52·5	6·691	1	2·929	11·8	46·7	9·70±0·11
40	3	193·9	−40·2	3·305	4	2·982	−76·5	277·2	—
42	4	6·7	−53·6	2·552	4	3·761	27·5	161·8	12·65±0·11
43	3	191·2	−21·9	2·179	6	2·977	−70·3	313·5	—
43A	4	224·3	+37·4	1·839	5	3·650	−25·0	302·0	—
44	3	191·7	−37·6	4·335	6	2·974	−77·3	288·7	10·30±0·13
45	3	18·2	+36·3	1·769	4	2·992	71·7	99·4	13·30±0·23
46	3	181·3	−38·3	1·445	5	2·943	−83·4	337·6	—
47	3	24·3	+65·8	4·270	5	2·982	62·5	24·6	10·90±0·22
48	4	185·5	−40·0	1·480	4	3·975	−82·6	301·6	13·10±0·34
48A	4	49·9	+57·1	2·285	4	3·765	47·5	51·7	—
48B	2	54·1	+46·0	5·000	4	1·900	42·5	61·5	—
49	2	335·9	+43·5	2·000	4	1·985	68·4	253·1	—
50	2	18·0	−22·9	8·000	4	1·888	46·5	138·1	—
51	2	15·4	+51·3	6·000	4	1·994	76·1	53·6	—
52	4	9·2	+41·1	1·672	2	3·870	80·6	104·2	15·0±0·20
53	2	351·9	+40·5	4·100	3	1·806	81·3	222·8	—
55	2	12·9	−06·9	9·000	4	1·868	56·3	140·7	—
56	2	44·9	+16·1	9·000	4	1·954	43·3	90·4	—
57	2	59·6	+5·2	3·000	4	1·999	27·9	87·1	—
59	2	2·9	+24·3	6·000	4	1·985	−74·6	153·6	—
60	2	172·7	−16·2	2·533	3	1·976	−69·1	8·9	—
61	2	243·2	+38·3	1·683	3	0·631	−11·4	290·2	—
62*	2	183·2	−34·9	1·386	3	1·994	−80·8	329·2	—
63	2	172·8	−40·4	1·553	3	1·989	−81·9	42·5	11·50±0·19
64	2	218·9	−42·6	2·244	3	1·998	−55·1	254·3	10·20±0·68
64A	2	180·2	−43·1	2·867	3	1·920	−87·0	344·1	—
(iv) Tenerife									
1*	6	285·4	−35·5	3·112	4	5·872	3·6	52·7	—
2	6	296·1	+12·7	6·507	4	3·792	26·0	250·4	—
3*	3	188·4	−60·3	6·899	4	2·977	−72·2	187·1	—
4	4	92·6	−6·1	1·319	4	3·528	−3·7	78·7	—
4A	4	22·1	+43·6	2·289	4	3·893	70·1	79·4	6·10±0·18
4B	4	5·1	+42·7	1·624	3	3·999	84·4	111·6	6·35±0·26
4C	4	13·7	+38·4	2·563	4	3·947	76·0	101·4	—
4D*	6	112·9	−69·4	1·053	4	5·948	−35·6	124·4	—
	4	331·4	+49·8	2·641	4	3·922	65·0	270·1	—
5*	6	146·5	−39·2	4·244	4	5·918	−59·2	73·9	5·35±0·11
6	4	33·5	+13·2	6·040	4	3·917	51·8	104·8	7·18±0·57
22*	3	86·2	+67·9	4·369	5	2·792	23·7	30·6	15·68±1·60
23*	4	291·0	−36·0	6·052	4	3·748	7·9	49·6	—
24*	6	166·7	−42·3	1·755	5	5·712	−77·6	63·5	—
25*	4	188·8	−56·8	1·335	4	3·931	−78·0	203·1	—

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Unit	N	D	I	J	(−E)	R	θ′	φ′	K:Ar(My)
(v) La Gomera									
1	2	33·6	+59·4	7·588	4	1·989	60·0	40·2	—
1A	3	287·1	−44·4	8·387	7	1·520	1·4	41·9	—
2	4	357·9	−8·4	2·388	4	2·826	57·5	166·6	14·60±0·67
3	6	355·1	+37·7	9·737	5	5·635	81·6	195·7	—
8	5	30·6	+43·1	9·127	4	4·918	62·6	72·1	—
8A	2	12·9	+41·5	7·552	4	1·997	77·7	90·4	—
9	2	322·9	+30·6	7·627	3	1·778	53·9	241·9	12·00±0·39
10	2	352·9	−4·4	1·713	3	1·980	58·9	176·8	—
12	2	16·3	+33·0	6·147	3	2·000	72·0	103·5	—
13	2	27·5	+22·8	4·115	3	1·998	59·6	99·7	—
14	2	354·0	+42·3	4·822	3	1·994	83·5	219·9	—
15	2	349·3	+40·2	1·150	2	1·980	79·0	226·8	—
16	2	356·2	+56·8	2·430	3	1·959	80·2	324·9	—
17	2	244·6	+68·6	1·317	3	1·999	8·0	308·4	8·64±0·14
18	2	228·1	−12·7	2·622	3	1·878	−39·7	269·9	±
19	2	355·3	+51·3	2·134	3	1·989	84·4	296·9	—
20	2	350·9	+48·3	1·871	3	1·999	82·0	293·7	—
21	2	187·6	−34·5	3·206	3	1·900	−78·5	304·0	—
22	2	348·4	+33·4	8·656	3	1·993	75·4	212·1	8·86±0·13
23	2	196·5	−23·3	1·306	2	1·999	−67·8	295·6	11·25±0·70
24	2	196·9	−24·5	3·925	3	1·910	−68·0	293·5	—
25	2	204·2	+4·6	3·304	3	1·925	−51·7	301·4	—
26	2	204·0	+4·6	3·200	3	1·997	−51·8	301·7	11·40±0·80
27	2	180·9	−39·9	1·488	3	1·898	−84·5	334·4	8·42±0·29
28	3	357·1	+16·4	2·877	4	2·454	70·0	171·1	—

Explanation of Table. Unit = number of sampled body on site location map (Figs 2–10). • = intrusive body (otherwise = extrusive). N = number of separate oriented samples. D and I = mean declination and inclination of remanent magnetism, following demagnetization treatment. D is in degrees east of north, and I is in degrees with respect to the horizontal (no sign = positive or below horizontal; negative = above horizontal); J = intensity of magnetization in emu g^{-1} ($-E$ = negative exponent of J); R = resultant vector, applying unit vector per specimen; θ' and ϕ' = latitude and longitude of virtual geomagnetic pole resulting from D and I at given sample sites. θ' is negative for southern latitudes, no sign for northern latitudes. ϕ' is in degrees east of north. K : Ar (My) = potassium–argon age determination for given unit (see *Abdel-Monem, Watkins & Gast* 1971, 1972; *Watkins & Abdel-Monem* 1971, for details).

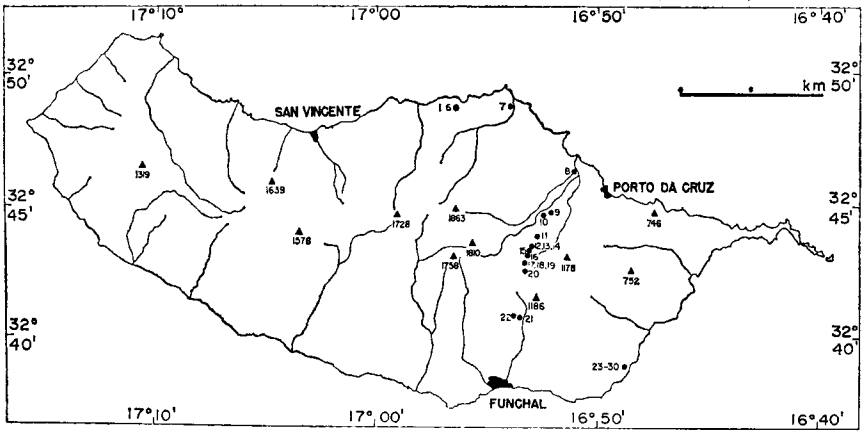


FIG. 2. Map of Madeira showing locations of sample sites (numbered 1–30, corresponding to Table 1). Elevation of selected topographic features given in metres above sea level.

Polished thin sections were manufactured for specimens from all bodies, for examination of the iron–titanium oxides using a Reichert Zetopan-Pol microscope: changes in the oxidation state and size of these minerals dominates in the magnetic property variation in basalts (Watkins & Haggerty 1967).

Full details of the experimental methods used in the K:Ar age measurements are given in Abdel-Monem *et al.* (1971, 1972) and Watkins & Abdel-Monem (1970).

Results

Figs 2–9 show the locations of each sample site. These maps can be compared with those in Abdel-Monem *et al.* (1971, 1972), which are the corresponding geological maps. If needed, precise site locations are available from the author on request.

The number of separate cores per body, all palaeomagnetic results, and statistical analysis for each body are given in Table 1, for each island. The data for those lavas of Brunhes age (t = 0–0·69 My) have been recently incorporated into a study of the palaeomagnetic secular variation in the Atlantic and Indian Ocean regions (Watkins *et al.* 1972). The K:Ar results are added to Table 1. The opaque mineralogical results have already been published (Ade-Hall & Watkins 1970).

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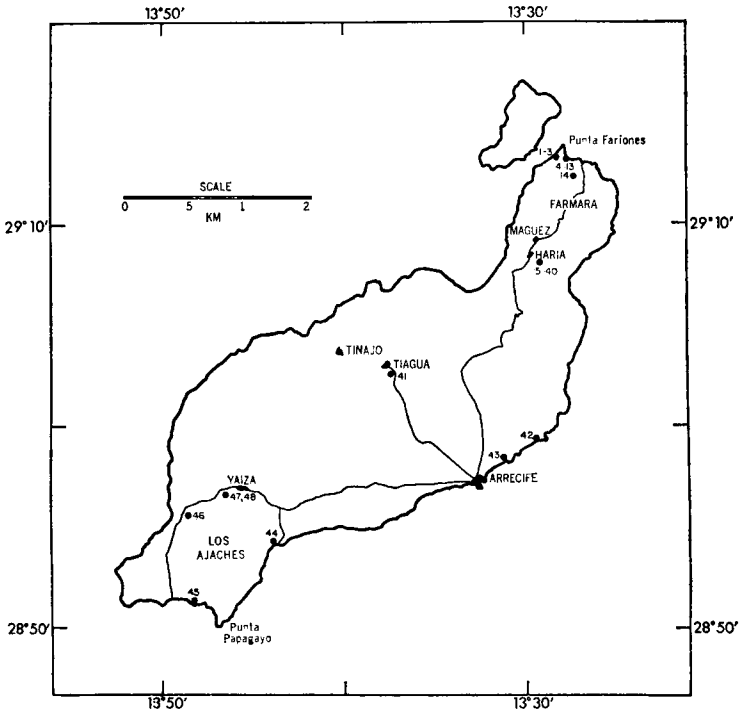


FIG. 3. Map of Lanzarote showing locations of sample sites (numbered 1–48, corresponding to Table 1).

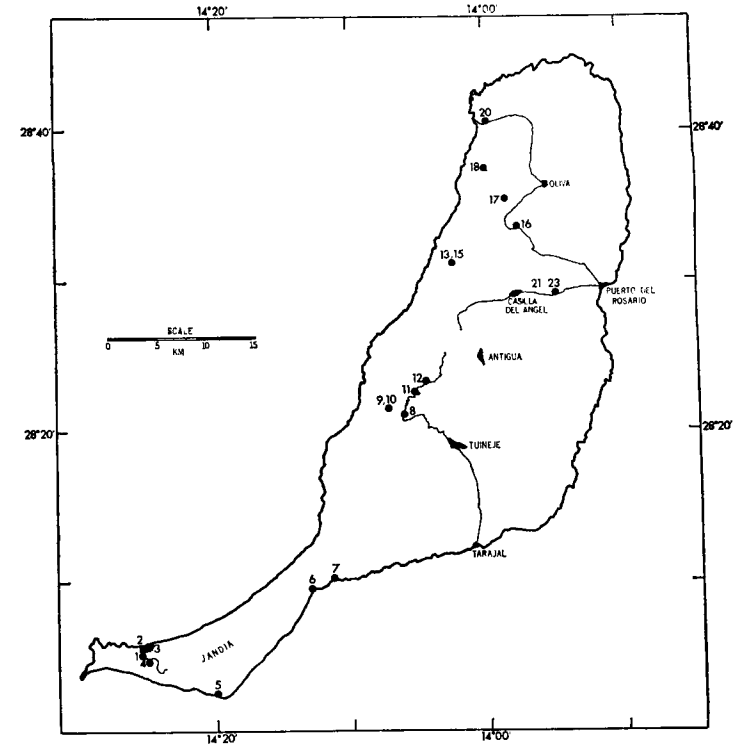


FIG. 4. Map of Fuerteventura showing locations of sample sites (numbered 1–23, corresponding to Table 1).

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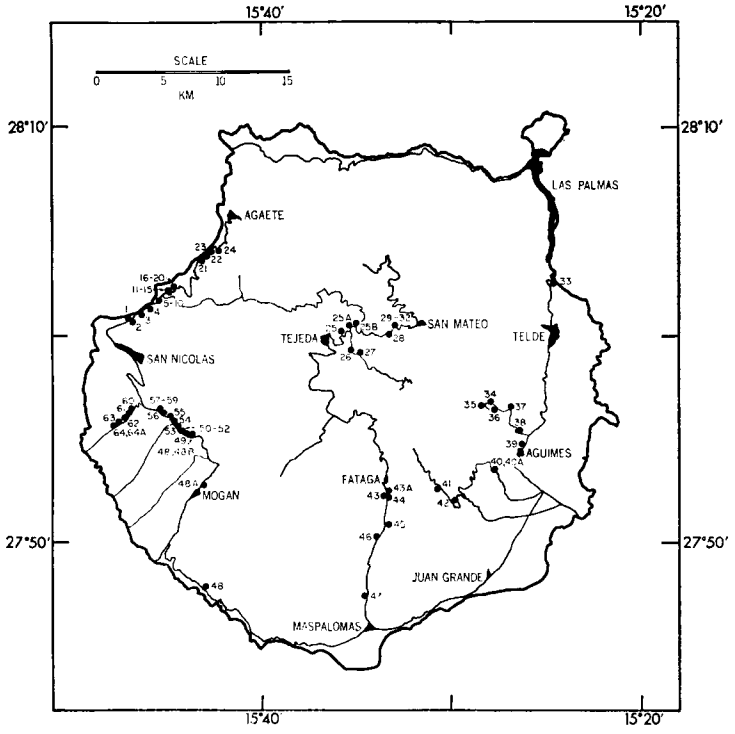


FIG. 5. Map of Gran Canaria showing location of sample sites (numbered 1–64A, corresponding to Table 1).

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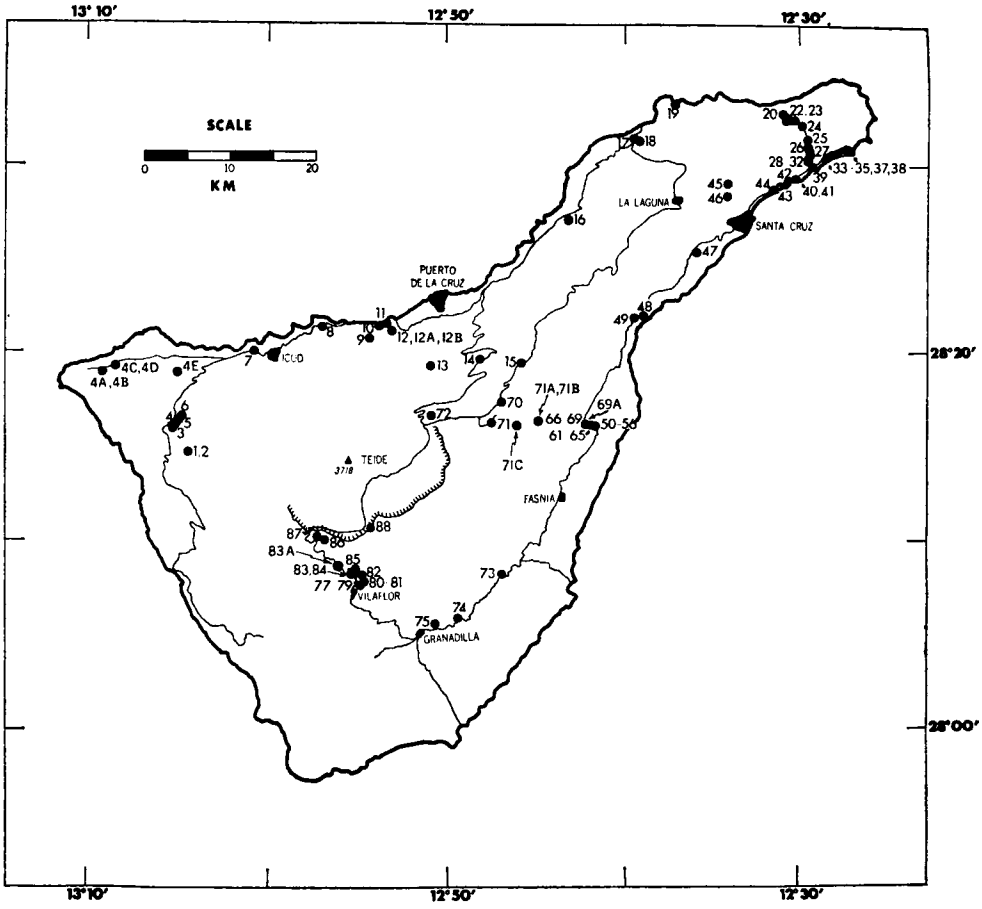


FIG. 6. Map of Tenerife showing location of sample sites (numbered 1–88, corresponding to Table 1). Height of Teide given in metres above sea level.

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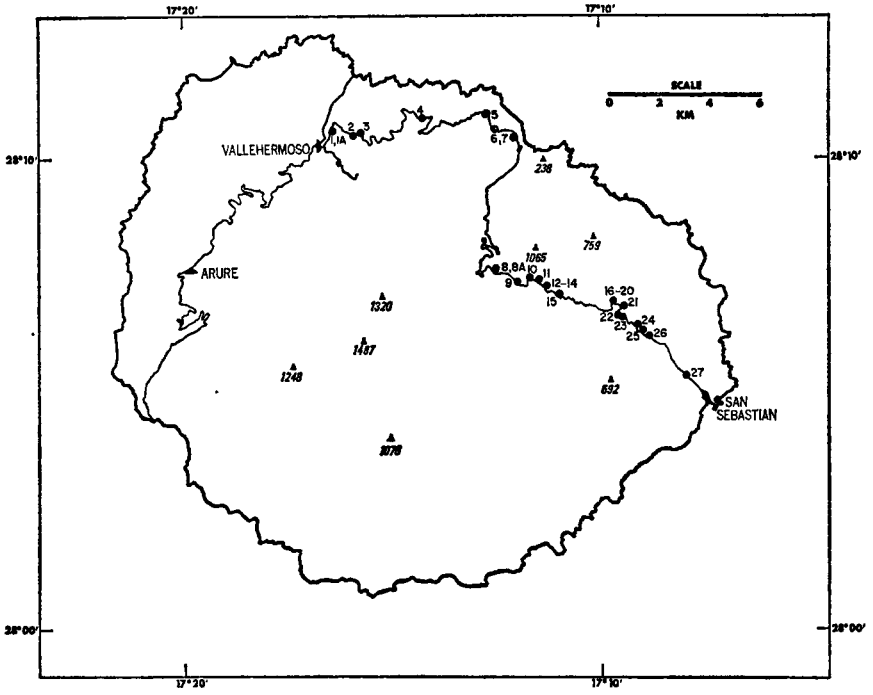


FIG. 7. Map of La Gomera showing location of sample sites (numbered 1–27, corresponding to Table 1). Height of selected topographic features given in metres above sea level.

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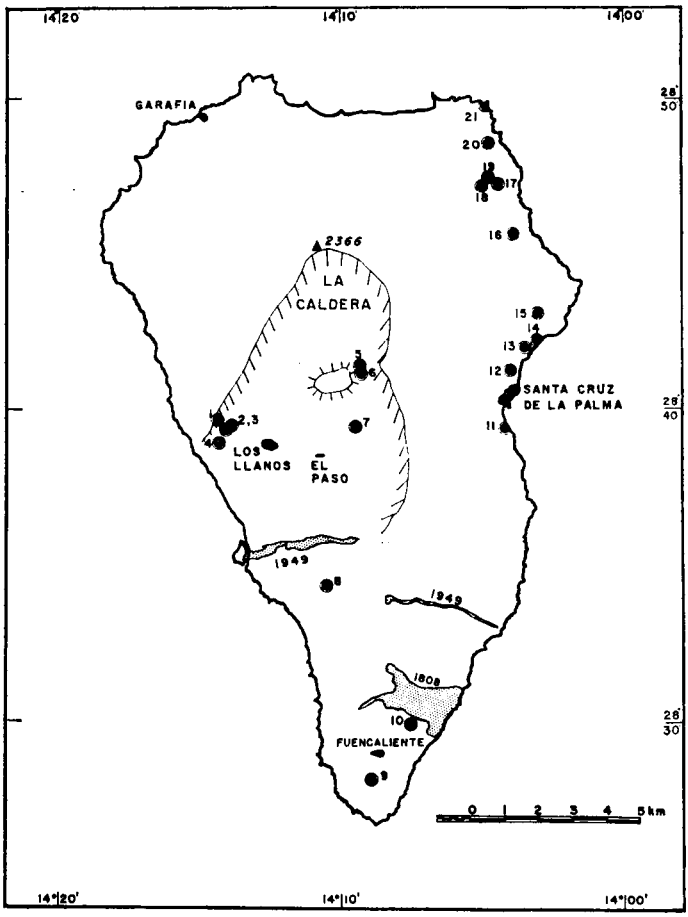


FIG. 8. Map of La Palma showing location of sample sites (numbered 1–21, corresponding to Table 1). Location of Recent lavas shown. Height of caldera rim given in metres above sea level.

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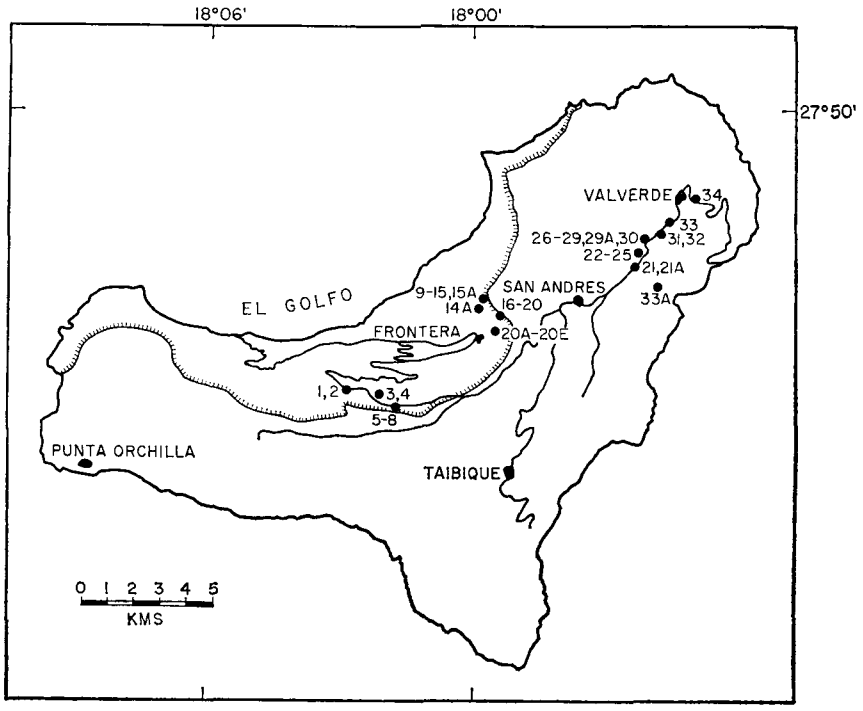


FIG. 9. Map of Hierro showing location of sample sites (numbered 1–34, corresponding to Table 1). Location of caldera rim shown.

Table 2(a)

Analysis of Palaeomagnetic data (directions): By island (for $B \geqslant 15$; θ' for each site $\geqslant 45°$)

Island	B	N	D	I	R	K	α_{95}	dp	dm	θ'	ϕ'
(a) Pleistocene											
Lanzarote	30	72	11·1	34·2	28·776	23·7	5·5	3·6	6·3	75·5	119·4
Tenerife	56	196	0·7	37·2	53·521	22·2	4·1	2·8	4·8	82·4	158·7
Hierro	35	87	8·5	47·1	32·849	15·8	6·3	5·3	8·1	82·4	65·3
La Palma	21	107	359·5	37·6	19·730	15·7	8·3	5·7	9·7	82·4	166·0
(b) Pliocene											
Gran Canaria	17	58	7·5	30·8	16·189	19·7	8·2	5·1	9·2	76·7	131·6
Tenerife	19	56	7·3	38·4	18·254	24·1	7·0	4·9	8·3	80·5	118·2
Madeira	22	50	1·5	45·9	21·550	46·6	4·6	3·7	5·9	84·5	149·1
(c) Miocene											
Lanzarote	15	53	18·8	39·4	14·472	26·5	7·6	5·4	9·1	71·8	93·8
Gran Canaria	43	102	4·9	40·2	39·686	12·7	6·4	4·6	7·7	83·8	122·6
La Gomera	22	54	5·2	31·2	20·082	10·9	9·8	6·1	11·0	77·8	138·5

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Table 2(b)

Analysis of Palaeomagnetic data (directions): By island (for $B \geqslant 15$; any value of θ')

Island	B	N	D	I	R	K	α_{95}	dp	dm	θ'	ϕ'
(a) Pleistocene											
Lanzarote	30	72	11·1	34·2	28·776	23·7	5·5	3·6	6·3	75·5	119·4
Tenerife	62	217	2·6	36·9	56·361	10·8	5·7	3·9	6·7	81·8	146·6
Hierro	35	87	8·5	47·1	32·849	15·8	6·3	5·3	8·1	82·4	65·3
La Palma	21	107	359·5	37·6	19·730	15·7	8·3	5·7	9·7	82·4	166·0
(b) Pliocene											
Gran Canaria	17	58	7·5	30·8	16·189	19·7	8·2	5·1	9·2	76·7	131·6
Tenerife	19	56	7·3	38·4	18·254	24·1	7·0	4·9	8·3	80·5	118·2
Madeira	22	50	1·5	45·9	21·550	46·6	4·6	3·7	5·9	84·5	149·1
(c) Miocene											
Lanzarote	17	67	20·7	40·9	15·791	13·2	10·2	7·5	12·4	70·6	88·7
Fuerteventura	16	65	349·7	36·7	11·407	3·3	24·4	16·6	28·5	77·7	217·8
Gran Canaria	53	127	9·7	39·4	43·273	5·3	9·4	6·7	11·2	79·6	105·0
Tenerife	15	68	336·6	42·5	10·411	3·1	26·5	20·2	32·7	68·7	249·0
La Gomera	25	61	4·2	31·3	20·898	5·8	13·1	8·2	14·7	78·2	143·0

Explanation of symbols used. B = total number of bodies; N = total number of separately oriented samples; D and I = mean declination and inclination of remanent magnetism in degrees east of north, and below horizontal respectively (obtained from unit vector per body); R = resultant vector (unit vector per body); $K = (N-1)/(N-R)$; α_{95} = semi-vertical angle of 95 per cent confidence cone; dp and dm = major and minor axes of 95 per cent confidence oval about mean pole position, in degrees along, and normal to, great circle from mean sample site location to mean pole position, respectively; θ' and ϕ' = latitude and longitude of mean virtual geomagnetic pole, obtained using unit vector per body. Note: data from all reversely magnetized bodies (θ' for each body negative) transposed to Northern Hemisphere.

Table 3

Analysis of Palaeomagnetic data (virtual geomagnetic poles): All islands combined

	B	N	R	K	α_{95}	θ'	ϕ'
(a) Pleistocene							
(i)	153	506	145·19	19·5	2·9	83·7	128·4
(ii)	162	533	150·04	13·5	3·0	82·8	119·8
(b) Pliocene							
(i)	71	214	68·4	26·8	3·2	82·6	128·4
(ii)	73	223	69·3	19·6	3·70	81·9	124·8
(c) Miocene							
(i)	99	291	92·9	16·0	3·5	81·9	114·4
(ii)	126	388	101·9	5·2	5·5	81·5	87·4
(d) Upper Tertiary (all data combined)							
(i)	322	1013	305·0	18·9	1·8	82·8	122·6
(ii)	359	1144	319·1	9·0	2·5	82·5	109·7

Explanation of symbols. As in Table 2, except that R, K, α_{95}, θ', and ϕ' result from analysis of virtual geomagnetic poles, applying unit vector per body. (i) is for data with θ' for each body (Table 1) less than 45° being rejected; and (ii) is for all data.

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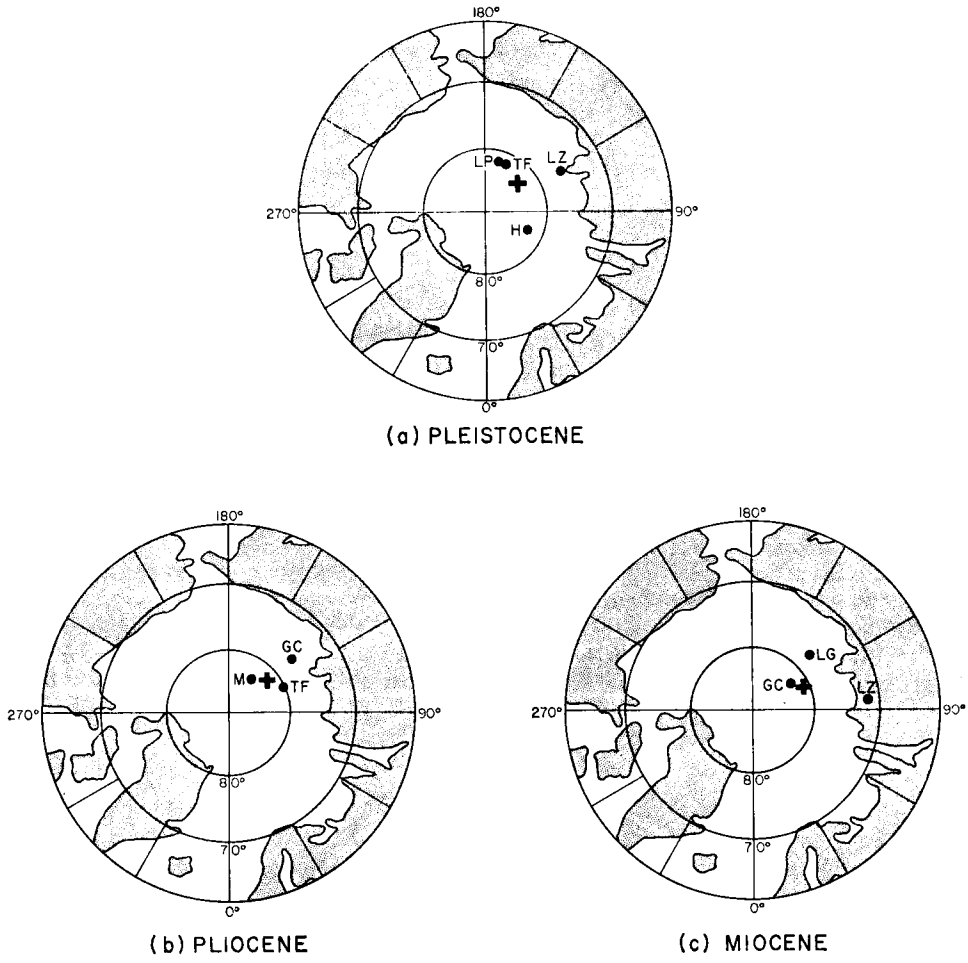


FIG. 10. Mean virtual geomagnetic poles for each island (minimum of 15 separate sites each with a virtual geomagnetic latitude greater than 45°: see Table 2), for each period, obtained using unit vector per site. LP = La Palma; TF = Tenerife; LZ = Lanzarote; H = Hierro; M = Madeira; GC = Gran Canaria; LG = La Gomera. Crosses in each diagram are the mean virtual geomagnetic poles for the respective periods (computed by averaging virtual geomagnetic poles for each site) using all data with virtual geomagnetic latitudes greater than 45°: see Table 3. Ninety five per cent confidence ovals for the mean virtual geomagnetic pole for each period are omitted for clarity. They have the following values: 2·9° for the Pleistocene; 3·2° for the Pliocene; and 3·5° for the Miocene.

The mean palaeomagnetic results for each island are given in Table 2(a), for groups containing at least 15 separate bodies for the Pleistocene, Pliocene, and Miocene periods. The age ranges of these periods are taken to be t = 0–1·6 My (see Kennett, Watkins & Vella 1971 for discussion); t = 1·6–5·1 My (Hays & Opdyke 1967) and t = 5·1–25·0 My, respectively. Data analysed are restricted to those bodies possessing a virtual geomagnetic pole (VGP) latitude higher than 45°. Using a minimum of 15 separate bodies (or points in time), and VGP latitudes greater than 45° will discriminate against the effect of non-axial dipole components in the mean data analyses. In Table 2(b), results of the same analyses are presented, without the VGP latitude acceptability criteria being applied. In Table 3, data from all islands

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are combined for statistical analysis for each period, both in the case of VGP > 45° and for all data. Fig. 10 shows the VGP for each island, by period, in those cases where the number of specimens is > 15; and the corresponding mean VGP for each period for all islands. These results supercede the earlier limited preliminary results of Watkins, Richardson & Mason (1966a, b).

Discussion

Within island data

The availability of a substantial number of K:Ar dates facilitates evaluation of prospects for finer definition and extension of the geomagnetic polarity history for the past 6–7 My by further work in the Canary Islands (For the period prior to about 7 My ago, the inherant precision limits of the K:Ar method will not allow confident extension of the time scale, as argued by Cox & Dalrymple (1967)).

Of the islands examined it is clear (Table 1, Figs 2–9) that Tenerife possesses the largest volume of Matuyama epoch ($t = 0{\cdot}69$–$2{\cdot}43$ My) material, although both La Palma and Madeira have large areas of Upper Matuyama outcrops. Gauss epoch volcanics ($t = 2{\cdot}43$–$3{\cdot}32$ My) are best developed on Gran Canaria, while the Gilbert epoch ($t = 3{\cdot}32$ to $\sim 5{\cdot}1$ My) materials are not common on any island. Only on Tenerife (on the westernmost peninsula in particular), and on Lanzarote do there appear to exist sequences of lavas which have recorded the geomagnetic polarities of the immediate pre-Gilbert epoch period ($t = 5{\cdot}1$–$7{\cdot}0$ My). Extensive field mapping which incorporate palaeomagnetic polarities is potentially very constructive for both local stratigraphic purposes, and for the reciprocal polarity time scale definition.

Between island data.

The mean virtual geomagnetic poles for each island (Table 2 and Fig. 10) are very similar to those from the Middle and Upper Tertiary of the Cape Verde Islands (Watkins, Richardson & Mason 1968), and the islands of the Gulf of Guinea, as summarized by Piper & Richardson (1972): no meaningful distinction exists between those poles. Because of this consistency, they provide a convincing record of the general direction of the Upper and Mid-Tertiary geomagnetic field for the central and north-eastern Atlantic region.

Combined results

The mean VGP for each period for all islands combined (Table 3 and Fig. 10) are very similar. They fall in the 90° to 180° East longitudinal quadrant which, when compared with the island locations, can be interpreted to be consistent with Wilson's (1970) offset dipole hypothesis: a northward offset of the main dipole along the spin axis results in a shallower geomagnetic inclination than that for a centred dipole, therefore yielding a computed palaeomagnetic co-latitude greater than the geographic co-latitude of the sites. Wilson (1971) has shown that an offset of 290 ± 90 km fits the best available Upper Tertiary ($t = 0$–25 My) palaeomagnetic results. At the latitude of the Canary Islands, this would mean that if such a dipole offset is ignored, and the conventional centred dipole equation is applied, then site palaeomagnetic directions due to a magnetic pole coinciding with the geographic pole will yield a virtual geomagnetic latitude at about $86{\cdot}5° \pm 1°$ (Fig. 11) in the opposite quadrant to the sites. Such a systematic distortion of the geomagnetic field could also be accomplished, however, by the superimposition of a minor axial dipole on the centred axial dipole, with the minor dipole being offset to the core–mantle boundary (Watkins 1972). Such a minor dipole would need to be only 3–4 per cent of the main dipole to create the same distortion in the Canary Island region as the hypothetical offset dipole of 290 km.

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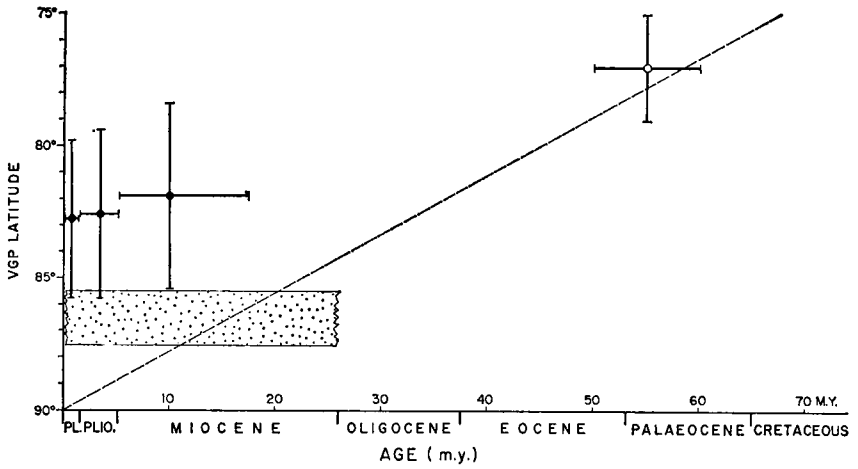


FIG. 11. Polar wandering diagram. The latitude of the virtual geomagnetic pole for the Pleistocene, Pliocene and Miocene periods from the Canary Islands and Madeira are plotted with their 95 per cent confidence intervals, at the mean ages. The age ranges of the three collections are added. The result for the Faeroe Islands (Tarling 1970) in the age range 50–60 My (Tarling & Gale 1968) is added. This is the only Lower Tertiary result from the Eurasian or African plate featuring over 70 separate sites and substantial K : Ar range definition. The diagonal line corresponds to an average polar wandering rate of 0·45° *of latitude* per million years. The dotted area extending from the Upper Pleistocene to $t = 25$ My corresponds to the latitudinal range created by an axial dipole offset 290±90 km (Wilson 1970) to the north, when the latitude is computed on a centred dipole assumption, in mid latitudes. See text for alternate source of this distortion of an axial dipole field, and for preferred interpretation of diagram.

Whatever their origin, the mean VGP for each period are clearly significantly different from the geographic pole (Figs 10 and 11). Perhaps more surprising is the almost identical positions of the VGP (Table 3). Because of the unique large volume and age control of the constituent data, this coincidence of VGP for the three geological periods must be considered to be probably realistic. The favoured interpretation of these results is the systematic long-period distortion of the geomagnetic field, by either an offset dipole or by an unequal co-axial dipole pair, as discussed above. This is much more feasible than a simple polar wandering model, since it would be necessary to invoke a greater than 5° per million years polar wandering rate, for the past 1·6 My to explain the results. Grommé & Vine (1972) believe that no true polar wander has occured since at least $t = 17$ My.

When the confidence limits are added to the mean VGP (Table 3, Fig. 11) no conclusions can be made concerning the polar wandering rate from the Pleistocene to the Miocene periods. But this could in principle be attempted by use of similar quality Lower Tertiary data from tectonically undisturbed parts of the African or Eurasian plates. The only Lower Tertiary result available which involves a large number (> 70) of separate points in time (sites) and substantial K:Ar age control is that due to the study of Tarling (1970) for the Faeroe Islands, which involved 253 separate lavas restricted to the period $t = 50$–60 My (Tarling & Gale 1968). When added to Fig. 10, this result is seen to be consistent with a polar wandering rate of 0·45° per million years directly away from the geographic pole which is compatible with several estimates of the average polar wandering rate for the Tertiary era (McElhinny & Wellman 1969). There is clearly no need to invoke an increase in the polar wandering rate at $t = 25$ My, as hypothesized by Irving & Robertson (1969) using Australian results, to explain the relationship between the data in Fig. 11: any change in the rate would appear to be more reasonably a decrease, unless it could be

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shown that long-term westward or eastward excursions were definitely superimposed on the latitudinal polar wandering.

Brock (1971) has reported the results of an extensive survey of the Gregory Rift of East Africa, by Raja (1968). While no details are presented, it is stated that the 120 separate sites involved have an age range of 0–7·0 My. The mean VGP is given as 148·7° East, 88·0° North.

While this pole is in the appropriate longitudinal segment, according to the above discussion, the latitude is significantly higher than those from the Canary Islands and Madeira, being (at the limits of the 95 per cent confidence oval) coincident with the present geographic pole. This result is at variance with almost all the Upper Tertiary results from the East Atlantic region, as summarized by Piper & Richardson (1972), and would not require any definite northward offset of an axial dipole. It is, however, consistent with the unequal co-axial dipole pair geomagnetic field model of Watkins (1972), one of the properties of which can be expressed as equivalent to a rapid decrease in apparent offset of a hypothetical axial dipole with increasing latitudinal separation from the north geographic pole. This model will be further discussed in a forthcoming publication. Whatever the explanation the virtual geomagnetic latitude of the Gregory Rift data, which Brock (1971) considers to be also indicative of no polar wandering during the past 7 My, it does not invalidate the preferred absence of an increase in the polar wandering rate at about $t = 25$ My.

The mechanism which has distorted the geomagnetic field away from that of an axial dipole for the Pleistocene through Miocene in the study region would appear not to have been effective for the rest of the Tertiary, according to the Faeroes result and average polar wander rate (Fig. 11). This is because migration of the VGP latitude away from that predicted by the average rate is clear for the Canary Island and Madeira data (Pleistocene to Miocene) but is not apparent for the Lower Tertiary result (Fig. 11). Despite the large number of sites and K:Ar results for each of the four VGP's involved, there are clearly insufficient results and knowledge of Lower Tertiary polar wandering (and, conceivably, Upper Tertiary rates) to make this suggestion more than a possibility, which is merely consistent with the limited data. A palaeomagnetic and K:Ar study of a large number of lavas of Oligocene age from Europe or Africa could clearly assist in resolving these ambiguities.

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